PQC DAILY STUDY SERIES

The Future of PQC: Beyond the First NIST Standards

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The Future of PQC: Beyond the First NIST Standards

As the world begins the transition to the first generation of NIST-standardized PQC algorithms (ML-KEM, ML-DSA, SLH-DSA), the cryptographic community is already looking ahead. The standardization process is not an endpoint but a milestone in a continuous journey to secure our digital infrastructure against future threats. Today, we explore what lies beyond these initial standards and how the landscape of post-quantum cryptography will continue to evolve.

The Need for Continuous Evolution

The current PQC standards are the result of years of intense scrutiny, but no algorithm is perfect, and the quantum threat itself is still theoretical. Several factors drive the need for ongoing research and development:

- **New Quantum Algorithms:** Shor's and Grover's algorithms are the best-known quantum threats, but new quantum algorithms could emerge, aiting the mathematical foundations of our current PQC standards.
- **Implementation Vulnerabilities:** Even with secure algorithm designs, vulnerabilities can be introduced during implementation (e.g., side-channel attacks). Research into "implementation-aware" cryptography and hardening techniques is crucial.
- **Performance and Efficiency:** The first PQC standards represent a trade-off between security and performance. Future algorithms may offer better efficiency, smaller key/signature sizes, or faster computation, making them more suitable for a wider range of applications, especially in resource-constrained environments like IoT.
- **Diversification of Cryptographic Assumptions:** The majority of the current lattice-based candidates (like Kyber and Dilithium) rely on similar mathematical problems. To avoid a cryptographic monoculture, NIST is already planning a future standardization round to encourage the development of algorithms based on different mathematical foundations.

What's Next on the PQC Roadmap?

NIST has outlined a path forward that includes a new call for proposals for public-key encryption and digital signature schemes, with a focus on algorithms that are not based on structured lattices. This is intended to diversify the portfolio of available PQC standards.

Key areas of active research include:

- **Isogeny-Based Cryptography:** Algorithms like SIKE (Supersingular Isogeny Key Encapsulation) were once promising candidates but were broken in 2022. However, research in this area continues, as it offers very small key sizes.
- **Hash-Based Signatures:** While SLH-DSA (SPHINCS+) is already standardized, it is a stateless hash-based signature scheme. Research into stateful schemes and other variations continues to offer different performance trade-offs.
- **Code-Based Cryptography:** Schemes like Classic McEliece offer high security but have large key sizes. Research is focused on reducing these overheads.
- **Symmetric-Key Cryptography:** The impact of quantum computers on symmetric-key algorithms like AES is less dramatic but still significant. Grover's algorithm effectively halves the key strength, which is why the transition to AES-256 is recommended.

Applicability to Internet Protocols

The evolution of PQC will directly impact internet protocols. Crypto-agility—the ability to easily swap out cryptographic algorithms—is more critical than ever. Future versions of TLS, SSH, and VPN protocols will need to be designed to accommodate new PQC standards with minimal disruption. Hybrid schemes, combining a classical and a post-quantum algorithm, will likely remain a key transition strategy, providing a safety net as new PQC algorithms are deployed and battle-tested in real-world environments.

Comparison of PQC Algorithm Families

Algorithm Family	Main Advantage	Main Disadvantage	Current NIST Standard
Lattice-Based	Strong security, good performance, relatively small key/signature sizes.	Relies on a single family of mathematical problems.	ML-KEM, ML-DSA
Hash-Based	Security is well-understood (based on the security of hash functions).	Signatures can be large, and some schemes are stateful.	SLH-DSA
Code-Based	Long history of security against both classical and quantum attacks.	Very large public key sizes.	N/A (was a finalist)
Isogeny-Based	Very small key sizes.	Currently broken, active area of research to fix.	N/A (was a finalist)
Multivariate	Small signatures.	Some schemes have been broken; complex security	N/A

Algorithm Family	Main Advantage	Main Disadvantage	Current NIST Standard
		analysis.	

> **Key Takeaway:** The PQC transition is a marathon, not a sprint. The first NIST standards are a critical first step, but the journey requires continuous research, development, and adaptation to stay ahead of emerging threats.

Daily Quiz

1. Why is cryptographic monoculture a concern in the context of PQC?
2. What does "crypto-agility" mean, and why is it important for internet protocols?
3. Which PQC algorithm family is known for its very large public key sizes but has a long history of security?
